Carrier board v1: soldering guide

Board 154 x 96 mm, bare copper (no printing). Every part goes on the TOP; every pin is soldered on the BOTTOM. Page 2: what the parts look like. Pages 3 and 4: both sides of the board. Page 5: if the holes are not plated.

Which side is which

1. Lay the board so the row of 4 large 5-pin footprints (module plugs J1-J4) is along the edge nearest you.
2. TOP (parts side): the long double row of 24 pads (Teensy socket) is at the LEFT, starting at the left edge; the short double row of 6 (DAC socket) is right of it, near the far edge. The 2 left mounting holes are lower (32 and 58 mm from the far edge) than the 2 right ones (10 and 45 mm).
3. If the Teensy rows are at the right instead, you are looking at the BOTTOM (solder side).
4. Check against page 3: the tracks you see on the top must match the red tracks there.

Before you start

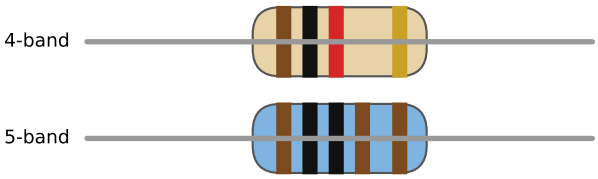
1. Look into a few holes. Copper-lined (shiny ring inside) = plated: solder on the bottom only. Bare fibreglass inside = NOT plated: also do page 5.
2. The board takes FEMALE headers (sockets). The Teensy and the DAC keep their male pins and plug in from the top. Never solder male pins into the board, and never solder the Teensy or DAC to it.
3. Multimeter in beep mode at hand. No Teensy, no DAC, no modules on the board while soldering.

Parts and order (lowest first; what they look like: next page)

#	Where	Part	How
1	R1-R4	4 x 1 kOhm resistor	either way round; holes 10 mm apart
2	R5-R8	4 x 1 MOhm resistor	either way round; holes 7.6 mm apart
3	D1-D4	4 x BAT85 diode (glass)	BAND towards the far edge (Teensy side): page 3
4	U1	2 x female header 1x24, 8.5 mm	Teensy socket, two rows
5	U2	2 x female header 1x6	DAC socket, two rows
6	J1-J4	4 x Phoenix MSTB 2,5/5-GF-5,08	plug opening towards the board edge

What the parts look like

Drawings, not to the same scale; real sizes in the text. German drawer labels in quotes. Bring this page and a multimeter: the resistor colours can be confusing, the meter is sure (1 kOhm reads 0.99-1.01 k, 1 MOhm 0.99-1.01 M).



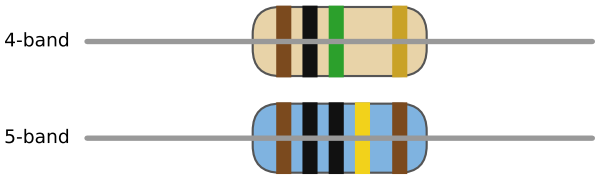
R1-R4: resistor 1 kOhm (x4)

"Widerstand 1 k" (1 kOhm, 1K0). Small cylinder, 6 mm long, 2.5 mm thick, a wire at each end.

4 bands (beige body): brown-black-RED-gold.

5 bands (blue body, 1 %): brown-black-black-BROWN-brown.

0.25 W (any size this small is fine).



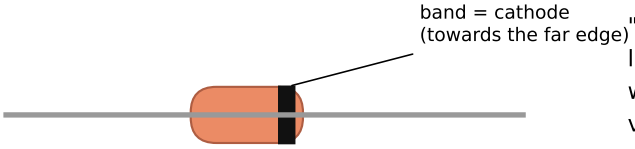
R5-R8: resistor 1 MOhm (x4)

"Widerstand 1 M" (1 MOhm, 1M0). Same size as the 1 k: only the colours differ.

4 bands: brown-black-GREEN-gold.

5 bands: brown-black-black-YELLOW-brown.

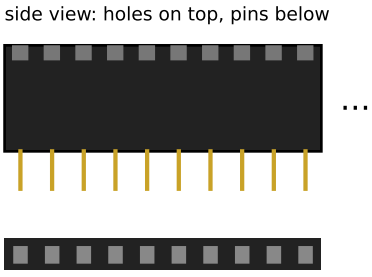
Do not mix up with 1 k: the 3rd (4-band) or 4th (5-band) colour is the difference.



D1-D4: diode BAT85 (x4)

"Schottky-Diode BAT85". Tiny glass body (3.8 mm long, 1.9 mm thick), orange-red and see-through, with a black band at one end = cathode. Printed text very small: use a magnifier.

Substitutes if there is no BAT85: BAT43, BAT46, BAT42 (same job). NOT a 1N4148 (that is the wrong type).

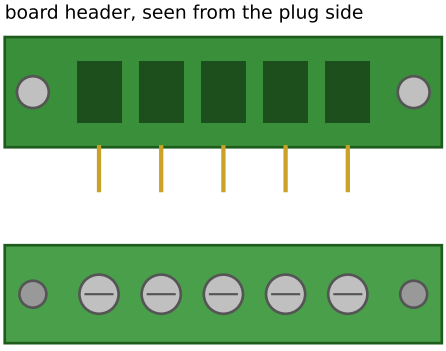


U1, U2: female headers (sockets)

"Buchsenleiste 2,54 mm, 1-reihig". Black plastic strip with a row of square holes on top, pins underneath. Need 2 x 24-pin (Teensy, 8.5 mm tall) and 2 x 6-pin (DAC).

Longer strips (1 x 40) are fine: cut them with side cutters THROUGH the next hole (you lose one pin), file flat.

NOT the "Stiftleiste" (male pin header: bare pins sticking up). Spare male strips only help hold the sockets straight while soldering.



J1-J4: Phoenix MSTB header (x4) + plug (x4)

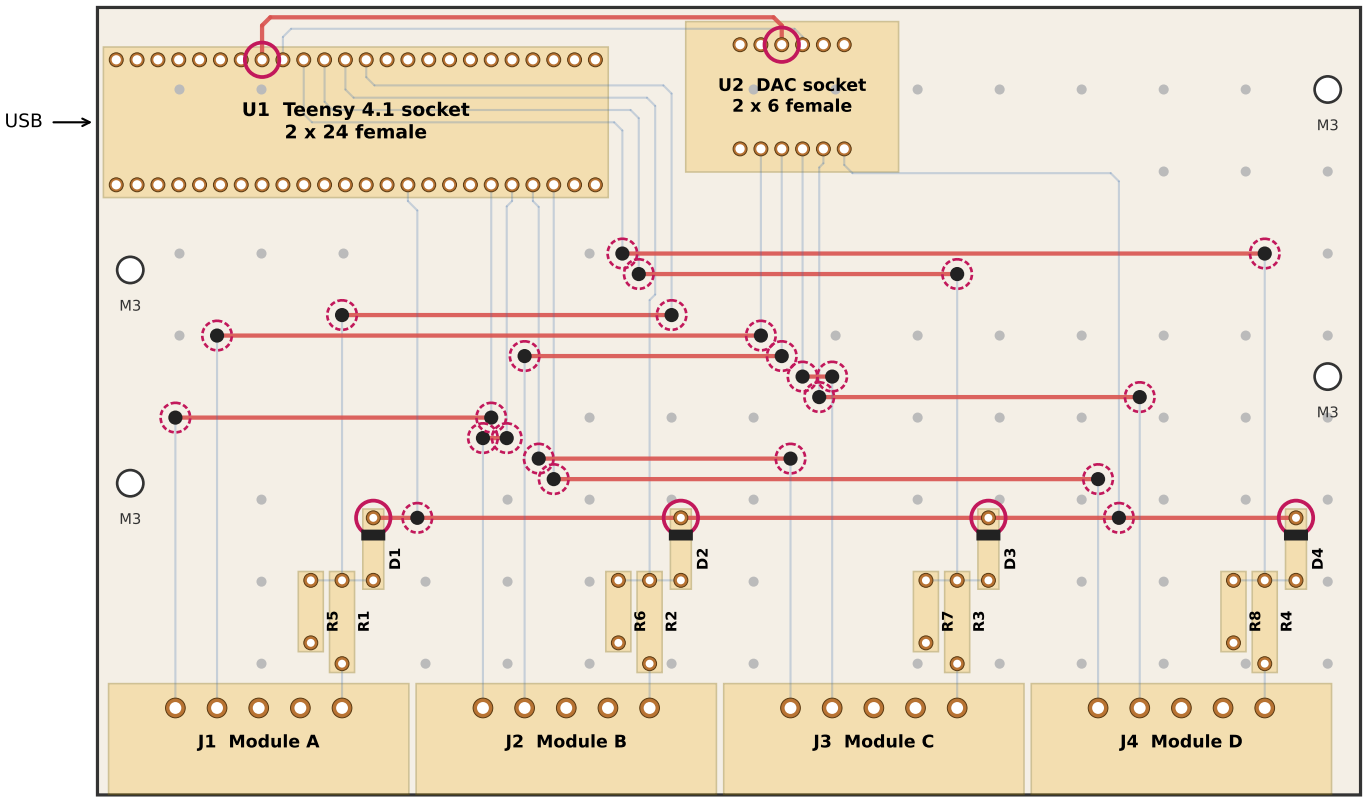
On the board: "Leiterplatten-Grundleiste" Phoenix MSTB 2,5/5-GF-5,08 (no. 1776537): green, 5 pin openings 5.08 mm apart, a threaded hole in a flange at each end, 5 pins underneath. The opening faces sideways (horizontal).

On each module cable: the plug MSTB 2,5/5-STF-5,08 (1778014): green, 5 screw terminals on top, wires in the side, a screw in each flange. Same family = they fit; other pitches (3.5 / 3.81 mm) do not.

No Phoenix in the makerspace? Order them (README): no substitute keeps the locking flanges.

3. TOP side: where every part goes

Seen from above, parts side up, module plugs at the bottom of the picture. Push each part in from this side; its pins come out on the bottom.



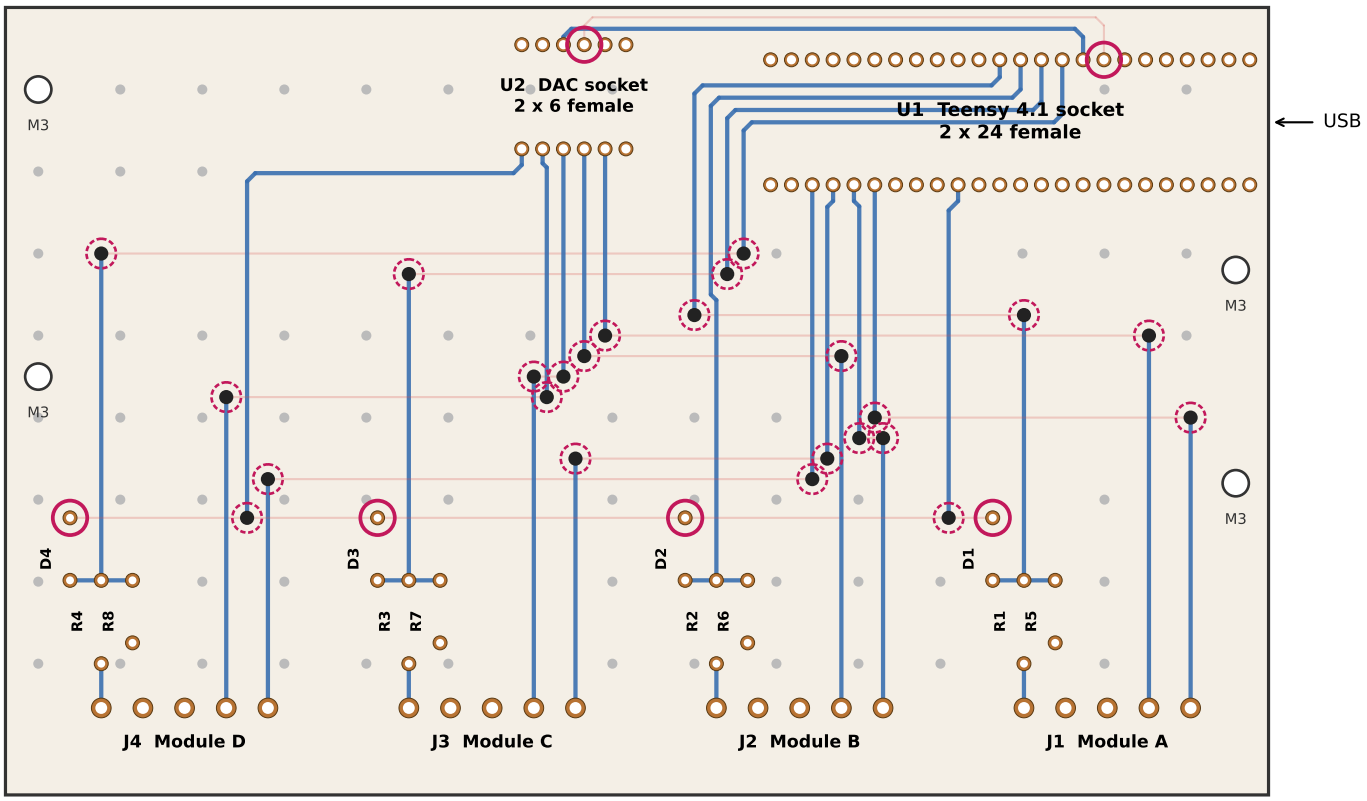
module plugs: this edge (cables plug in here)

where the part body sits
top copper track (on this side)
bottom track (other side, seen through)
pad with hole (one pin each)
via: track changes side here (small grey dots: ground vias, nothing to do)
circled: needs solder ON TOP too, if the holes are not plated (page 5)

1. Resistors and diodes: bend the leads to the hole spacing, push them in until the body lies on the board, bend the leads out a little underneath so they stay, solder, cut.
2. Diodes D1-D4: the black band (black bar in the picture) goes to the hole nearer the far edge (the 3.3 V line). Wrong way round, it shorts the force signal to 3.3 V.
3. Sockets: plug the female headers onto the Teensy (and the DAC) or onto spare male pin strips, set them in, solder one pin at each end, check they sit flat and straight, then solder the rest.
4. MSTB plugs J1-J4: the opening where the cable plug goes in faces the near board edge; the flange screws are at its ends.

4. BOTTOM side: where you solder

The board turned over (left and right swap: the Teensy is now on the RIGHT). Every pad here gets one solder joint: heat pad and pin together 2-3 s, feed solder until it forms a small cone, cut the leads ~1 mm.



module plugs: this edge (cables plug in here)

- bottom copper track (on this side)
- top track (other side, seen through)
- pad with hole (one pin each)
- via: track changes side here (small grey dots: ground vias, nothing to do)
- circled: needs solder ON TOP too, if the holes are not plated (page 5)

1. Small joints, no bridges: socket pins are only 2.54 mm apart. After each row, look at it against light.
2. Ground pads sit in the copper fill and take longer to heat: hold the iron 1-2 s more.
3. Circled pads and vias: if the holes are not plated, they also need the top-side work on page 5.

5. Only if the holes are NOT plated

In a plated board the copper inside each hole joins the top and bottom. Without it, a joint on the bottom does not reach a top track. Two places need extra work (circled in pink on pages 3 and 4):

A. 6 pads with a track only on the TOP: solder them on the top as well

- D2 pad 1 (diode, band end): +3V3
- U1 pad 41 (Teensy socket, Teensy pin 19): SCL
- D1 pad 1 (diode, band end): +3V3
- U2 pad 3 (DAC socket, its SCL pin): SCL
- D4 pad 1 (diode, band end): +3V3
- D3 pad 1 (diode, band end): +3V3

Diodes D1-D4: the band-end lead is reachable next to the body: solder it on the top as well as underneath (the top track joins the 4 diodes to 3.3 V). SCL (U1 pad 41 = Teensy pin 19, far row, 8th from the left; U2 pad 3 = DAC far row, 3rd from the left): the top track runs only between these two pins and the socket plastic covers it. Easiest: ignore that track and solder a short insulated wire on the BOTTOM from the one pin to the other.
Ground needs nothing: every ground pin reaches the bottom copper fill, which is one piece (checked on the board file, 2026-10-26).

B. 24 signal vias: a short wire through each, soldered on both sides

A via is where a track goes from one side to the other. Do them right after the resistors and diodes (no via sits under a part), before the sockets: push a cut-off resistor or diode lead (or thin solid wire, max 0.7 mm) through, solder on top and bottom, cut both ends short. The 12 parts give exactly 24 offcuts; keep them. Grey vias (ground): leave them.

- +3V3: (125, 62) mm, (39, 62) mm
- EN_A: (48, 50) mm, (10, 50) mm
- EN_B: (50, 52) mm, (47, 52) mm
- EN_C: (54, 55) mm, (84, 55) mm
- EN_D: (122, 58) mm, (56, 58) mm
- F_A: (30, 38) mm, (70, 38) mm
- F_C: (66, 32) mm, (105, 32) mm
- F_D: (142, 30) mm, (64, 30) mm
- SP_A: (15, 40) mm, (81, 40) mm
- SP_B: (52, 42) mm, (83, 42) mm
- SP_C: (90, 45) mm, (86, 45) mm
- SP_D: (127, 48) mm, (88, 48) mm

Positions in mm from the top-left corner seen from the top (USB edge left, plug edge at the bottom). Check each with the multimeter afterwards: beep from the pad on one side to the track it should reach on the other.

6. Check before plugging anything in

Multimeter on beep. No Teensy, no DAC, no modules (README 'Assembly and first power-up').

1. 3V3 to GND must NOT beep: Teensy socket 3V3 pin to any J pin 3 or 4.
2. Each plug J1-J4: pins 3 and 4 to a Teensy socket GND pin must beep.
3. J1 pin 1 to Teensy socket pin 27 must beep; J2 pin 1 to 28, J3 to 29, J4 to 30 (module enable).
4. DAC socket SCL to Teensy pin 19 and SDA to pin 18 must beep (these use the top track and a via: page 5).
5. Each J pin 2 (setpoint) to its DAC output pin must beep (A-D: through vias).
6. Then plug in the DAC and the Teensy (USB at the board edge), USB only: bumpem info --port <PORT> must answer.
7. One module at a time, 48 V off: plug in, then scripts/open_loop_test.sh for that module.